

HAZOP Analysis: Three-Stage Claus Sulfur Recovery Unit

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Simulation Basis: Aspen Plus V14 SR-POLAR Property Package

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Scope and Methodology

This HAZOP covers the three-stage Claus sulfur recovery unit modeled in Aspen Plus V14. The analysis is based on the process flow diagram and simulation output not a formal P&ID. Where plant-specific instrument data is absent, safeguards are taken from standard industrial Claus practice.

Five nodes, in process order:

1. Reaction Furnace (RF) + Waste Heat Boiler
2. First Condenser (C)
3. Reheater 1 + Catalytic Converter 1 ($H_1 \rightarrow CR_1$)
4. Reheaters 2 & 3 + Catalytic Converters 2 & 3 ($H_2 \rightarrow CR_2$, $H_3 \rightarrow CR_3$)
5. Tail Gas / OFFGAS System

Guide words: No/None, More, Less, Reverse, Other Than, As Well As, Part Of

Parameters: Flow, Temperature, Pressure, Composition, Level (where applicable)

Node 1: Reaction Furnace (RF) + Waste Heat Boiler

Design Intent: Combust approximately one-third of the H_2S feed with sub-stoichiometric air to produce SO_2 , achieving an $H_2S:SO_2$ molar ratio of 2:1 at furnace outlet. Reach adiabatic equilibrium at $995.6^\circ C$, 1.5 bar. The waste heat boiler cools the hot gas to $\sim 300^\circ C$ before the first condenser.

Normal Operating Conditions:

- Acid gas: 171.1 mol/s, $218^\circ C$, 1.77 bar (36.04 mol% H_2S , 53.16 mol% CO_2)
- Air: 181.5 mol/s, $220^\circ C$, 1.68 bar
- Fuel gas: 3.25 mol/s, $40^\circ C$, 6.0 bar
- Furnace outlet temp: $995.6^\circ C$
- H_2S in HOT-GAS: 54.58 kmol/hr; SO_2 : 30.49 kmol/hr; S_2 : 66.52 kmol/hr

#	Parameter / Guide Word	Deviation	Possible Causes	Consequences	Safeguards (Industry)	Recommended Action
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1.1	Air Flow No/None	No air to furnace	Air blower failure; blocked air line; control valve failure closed	No SO ₂ produced; H ₂ S:SO ₂ ratio collapses; unburned H ₂ S passes downstream; potential furnace flame-out; catastrophic H ₂ S release	Low-flow alarm on air line; flame detector on furnace burner; automatic fuel gas and acid gas trip on loss of combustion	Install air flow low-low trip (FFLL) interlocked with acid gas feed shutoff. Flame detector mandatory.
1.2	Air Flow More	Excess air to furnace	Air ratio controller failure high; control valve failure open	Excess O ₂ reacts with H ₂ S to form SO ₃ ; catalyst sulfation in downstream beds; risk of SO ₂ overproduction; H ₂ S:SO ₂ ratio goes below 2:1, reducing downstream conversion	High-flow alarm on air line; O ₂ analyzer on furnace outlet	Add O ₂ analyzer (AIC) on HOT- GAS stream with high O ₂ alarm. Air flow ratio controller (FFIC) relative to acid gas mandatory.
1.3	Air Flow Less	Low air to furnace	Partial blower failure; filter fouling; control valve partially closed	Sub- stoichiometric combustion worsens; H ₂ S:SO ₂ ratio exceeds 2:1; increased H ₂ S slip to condensers; incomplete combustion; CO formation increases (already 24.42 kmol/hr at design)	Flow indicator controller on air; ratio controller vs acid gas	Ratio controller FFIC with low- flow alarm. Regular air filter inspection.
1.4	Acid Gas Flow No/None	No acid gas feed	Upstream amine regenerator trip; feed block valve closed;	Furnace flame may be sustained by fuel gas only; air:fuel ratio shifts dangerously; risk	Feed flow low alarm; automatic air trip on loss of acid gas	Interlock: acid gas FFLL trips air supply and fuel gas simultaneously.

			pipeline rupture	of explosive atmosphere if air continues without H ₂ S combustion		
1.5	Acid Gas Flow More	Excess acid gas	Upstream pressure surge; control valve failure open	More H ₂ S than air can combust; H ₂ S:SO ₂ ratio increases well above 2:1; increased H ₂ S carrythrough; potential flame instability	High flow alarm on acid gas; ratio controller	FFIC ratio control. High-flow alarm with operator alert. Consider FFHL interlock.
1.6	Acid Gas Comp. Other Than	High H ₂ O content spike	Upstream amine system upset; regenerator flooding	Lower flame temperature; increased COS/CS ₂ formation; potential furnace quench if severe	Feed composition analyzer (H ₂ S, CO ₂ , H ₂ O online)	Online feed gas analyzer recommended. Low temperature alarm on furnace outlet (TAL).
1.7	Acid Gas Comp. Other Than	H ₂ S content drops (lean acid gas)	Upstream process change; solvent lean loading shift	Flame temperature drops; potential flame extinction below ~900°C; H ₂ S:SO ₂ ratio control lost	Temperature indicator on furnace; fuel gas supplementation system	Fuel gas flow controller (FIC) with automatic boost on low furnace temperature. TAL on RF outlet at 900°C.
1.8	Temperature More	Furnace overtemperature (>1400°C)	Excess fuel gas; high H ₂ S content spike; air/acid gas ratio upset	Refractory damage; waste heat boiler tube failure; downstream metallurgy stress	High temperature alarm and trip (TAHL) on furnace; refractory thermocouple monitoring	TAHH trip on furnace at 1450°C interlocked with fuel gas and acid gas shutoff.
1.9	Temperature Less	Low furnace temp (<900°C)	Lean acid gas; low fuel gas; high CO ₂ dilution (already	Incomplete combustion; higher COS/CS ₂ formation; flame instability; liquid	Low temperature alarm (TAL); fuel gas makeup	TAL at 900°C triggers fuel gas boost. TALL at 850°C trips acid

			present at 53.16 mol%)	sulfur condensation in furnace		gas feed.
1.10	Fuel Gas Flow More	Excess fuel gas	FIC failure high; control valve failure open	Furnace overtemperature; excess CO in HOT-GAS (already 24.42 kmol/hr at design); increased carbon deposition risk	High flow alarm on fuel gas; temperature controller on furnace	FFHL on fuel gas interlock with temperature.
1.11	Fuel Gas Comp. Other Than	Higher hydrocarbon content (C ₃ +)	Pipeline supply change	Higher flame temperature; carbon deposition risk; catalyst contamination downstream	Fuel gas composition analyzer	Periodic fuel gas composition check.
1.12	Pressure More	High pressure in furnace	Downstream blockage; condenser fouling; control valve failure closed on outlet	Reverse flow risk into feed lines; furnace casing stress	Pressure indicator on furnace outlet; PSV on furnace	PSV on furnace casing mandatory. High pressure alarm (PAHL) at 1.8 bar.
1.13	Pressure Less	Low furnace pressure	Leak in furnace casing or refractory; upstream pressure drop	Air ingress into process; risk of explosive mixture; H ₂ S/SO ₂ leak to atmosphere	Pressure indicator; H ₂ S gas detectors around furnace	H ₂ S detectors (fixed) around furnace perimeter. Low pressure alarm (PALL).
1.14	WHB Cooling No/None	Loss of cooling in WHB	Cooling water pump failure; tube rupture	HOT-GAS exits WHB above 300°C; first condenser thermal overload; liquid sulfur re-	Cooling water flow indicator; WHB outlet temperature indicator	TAH on WHB outlet (>350°C) trips acid gas feed. Cooling water FFLL alarm.

				vaporization		
1.15	WHB Cooling Other Than	Tube rupture in WHB	Thermal fatigue; water hammer	Water injection into process gas; steam generation inside process; downstream H ₂ S/water slug	Differential pressure monitoring across WHB	ΔP alarm across WHB. Steam drum level monitoring. Emergency isolation of WHB.

Node 2: First Condenser (C)

Design Intent: Cool HOT-GAS from 995.6°C to 130°C at 1.5 bar. Condense and drain liquid sulfur (predominantly S₂). Vapor fraction leaving = 0.946. Heat duty = 3.24×10⁶ cal/s. Liquid sulfur (LS stream): 66.34 kmol/hr S₂, 2.63 kmol/hr H₂O.

#	Parameter / Guide Word	Deviation	Possible Causes	Consequences	Safeguards (Industry)	Recommended Action
2.1	Temperature Less	Condenser outlet below 130°C	Excess cooling water flow; cooling medium temperature drop	Liquid sulfur approaches solidification point (119°C); sulfur blockage in drain; condenser plugging	Temperature indicator controller (TIC) on condenser outlet; cooling water control valve	TIC with low alarm at 125°C. TALL at 120°C trips cooling water supply. Heat tracing on sulfur drain lines mandatory.
2.2	Temperature More	Condenser outlet above 130°C	Low cooling water flow; fouling on tube side; cooling water supply failure	Reduced sulfur condensation; more sulfur vapor carried to CR1; catalyst bed loading increases; sulfur dew point issues downstream	High temperature alarm (TAH) on condenser outlet	TAH at 140°C. Investigate tube fouling if persistent high temp at normal cooling flow.
2.3	Cooling Flow No/None	No cooling water	Pump failure; supply line rupture; utility failure	Condenser acts as pass-through; HOT-GAS enters CR1 at near- furnace temperature	Cooling water flow indicator; low flow alarm	FFLL on cooling water trips acid gas feed after 2-minute delay. Redundant cooling water pump.

				($>900^{\circ}\text{C}$); catalyst sintering; converter destruction		
2.4	Sulfur Level High	Liquid sulfur accumulates	Drain valve failure closed; sulfur solidification in drain line; drain pump failure	Sulfur floods condenser; sulfur carryover into PROC-G1 vapor stream; downstream catalyst contamination	Level indicator on liquid sulfur sump; high level alarm (LAH)	LAH with automatic drain valve open signal. Heat tracing on all sulfur drain lines (maintain $>130^{\circ}\text{C}$).
2.5	Sulfur Level Low/None	No liquid sulfur collecting	Upstream process upset; very low H_2S feed	May indicate upstream problem not hazardous in isolation but signals process deviation	Level indicator	Level low alarm as diagnostic indicator.
2.6	Pressure More	High pressure in condenser	Downstream blockage; control valve failure; PROC-G1 line restriction	Back-pressure on furnace; reduced condenser efficiency	Pressure indicator; PSV	PSV on condenser vessel. PAHL alarm.
2.7	Sulfur Drain No Flow	Drain valve stuck closed or sulfur solidified	Drain valve failure; heat trace failure; temperature drop below 119°C	Liquid sulfur accumulates; level rises; sulfur entrainment into vapor outlet	Level alarm; heat trace monitoring	Redundant heat trace on sulfur drain. LAH interlock with drain override valve.
2.8	Water Content More	Higher H_2O in liquid sulfur	Upstream H_2O spike from acid gas or WHB tube leak	H_2S dissolved in liquid sulfur water phase; H_2S release during sulfur storage/handling	LS stream composition monitoring	H_2S detector at sulfur pit/storage. Degassing of liquid sulfur before storage is standard practice.

Node 3: Reheater 1 + Catalytic Converter 1 (H1 → CR1)

Design Intent: H1 reheats PROC-G1H from 130°C to 260°C (heat duty 3.71×10^5 cal/s). CR1 converts H_2S + SO_2 catalytically to S_8 and H_2O at adiabatic equilibrium; outlet temperature 307.5°C (47.5°C exotherm). H_2S drops from 54.52 to 19.83 kmol/hr (15.7 percentage points of recovery). COS drops 98.9% in this stage.

#	Parameter / Guide Word	Deviation	Possible Causes	Consequences	Safeguards (Industry)	Recommended Action
3.1	Reheater Temp More	PROC-G1H temp above 260°C	Steam supply pressure high; TIC failure high; control valve failure open	CR1 inlet too hot; equilibrium shifts toward reactants; lower conversion; potential catalyst sintering above 400°C	TIC on reheater outlet; steam pressure control	TAHL on H1 outlet at 300°C. Steam supply pressure regulator upstream of H1.
3.2	Reheater Temp Less	PROC-G1H temp below 260°C	Steam supply failure; TIC failure low; heat exchanger fouling	CR1 inlet too cold; catalyst inactive below ~200°C; H_2S and SO_2 pass through unconverted; downstream recovery loss	TAL on H1 outlet	TAL at 240°C. TALL at 200°C with operator alarm consider acid gas feed rate reduction.
3.3	Steam Supply No/None	No steam to H1	Boiler trip; steam header failure; isolation valve closed	No reheating; PROC-G1 enters CR1 at 130°C; catalyst inactive; no conversion in CR1; all downstream recovery depends on CR2 and CR3 alone	Steam flow indicator; low alarm	FFLL on steam supply to H1. Alarm to control room. If steam lost to all three reheaters simultaneously, consider unit shutdown.
3.4	CR1 Temp More	Converter outlet above 320°C	Excessive $\text{H}_2\text{S}/\text{SO}_2$ feed; inlet temperature too high; unusually high	Approaches catalyst sintering range (>350°C for alumina); conversion actually	TAH on CR1 outlet	TAHL at 330°C. Review $\text{H}_2\text{S}:\text{SO}_2$ ratio if TAH activates.

			conversion exotherm	decreases as equilibrium shifts back		
3.5	CR1 Temp Less	Low temp rise across CR1 (<20°C)	Low H ₂ S/SO ₂ concentration entering CR1; catalyst deactivation; catalyst bed channeling	Low conversion in CR1; increased load on CR2 and CR3; overall recovery risk	Temperature differential indicator across CR1	Monitor ΔT across CR1. Significant drop in ΔT from design (47.5°C) indicates catalyst degradation schedule bed inspection.
3.6	Feed Comp. Other Than	Sulfur liquid droplets in PROC-G1	Condenser C level high; sulfur carryover	Catalyst bed fouling by liquid sulfur; pore blockage; permanent catalyst deactivation	Mist eliminator on condenser vapor outlet	Install mist eliminator/demister pad on C vapor outlet. LAH on C sump.
3.7	Feed Comp. Other Than	High COS entering CR1	High CO ₂ in acid gas (already 53.16 mol%); furnace temperature low	COS normally 98.9% hydrolyzed in CR1 elevated COS inlet increases load but Keq is large (4.15×10^6 at 307.5°C); risk increases only if catalyst is deactivated	COS analyzer on PROC-G1 (optional)	Online COS monitor recommended if acid gas CO ₂ content is variable. First bed TiO ₂ catalyst preferred over alumina for high CO ₂ feed.
3.8	H ₂ S:SO ₂ Ratio Other Than	Ratio deviates from 2:1	Air ratio controller upset; furnace combustion instability	Off-ratio feed to CR1; conversion efficiency drops; tail gas composition shifts; potential SO ₂ excess or H ₂ S excess in tail gas	H ₂ S:SO ₂ online analyzer on PROC-G1	H ₂ S/SO ₂ ratio analyzer on process gas entering CR1. Feedback to air flow ratio controller on furnace. This is the most critical composition control point in the unit.
3.9	Pressure More	High pressure across CR1	Catalyst bed plugging by sulfur or carbon; high	Increased pressure drop; gas maldistribution;	Differential pressure indicator	ΔP indicator across CR1 with high alarm. Rising ΔP trend = early warning of bed

gas velocity channeling across CR1 fouling.
through bed

Node 4: Reheaters 2 & 3 + Catalytic Converters 2 & 3

Design Intent: H2 reheats to 220°C (duty 2.50×10^5 cal/s); CR2 outlet 236.6°C (16.6°C exotherm, 6.0 pp recovery). H3 reheats to 190°C (duty 1.66×10^5 cal/s); CR3 outlet 194.3°C (4.3°C exotherm, 1.6 pp recovery). H₂S drops from 19.83 → 6.36 → 2.89 kmol/hr. C2 and C3 condensers return vapor-only streams in simulation (small sulfur quantities expected in real plant).

#	Parameter / Guide Word	Deviation	Possible Causes	Consequences	Safeguards (Industry)	Recommended Action
4.1	CR2 Inlet Temp More	Above 220°C into CR2	H2 TIC failure high; steam overpressure	Equilibrium shifts toward reactants at higher temperature; conversion drops in CR2; Keq for Claus reaction decreases sharply with rising temperature	TIC on H2 outlet	TAHL at 240°C on H2 outlet.
4.2	CR2 Inlet Temp Less	Below 220°C into CR2	Steam supply low; H2 fouling	Lower catalyst activity; suboptimal conversion; increased load on CR3	TAL on H2 outlet	TAL at 200°C. Note: CR3 operates at 190°C if CR2 drops near this temperature, the two stages effectively merge with very little benefit from CR2.
4.3	CR3 Inlet Temp Less	Below 190°C into CR3	H3 steam failure; heat loss	Catalyst operates below design; marginal conversion gain further reduced; at 119°C sulfur solidifies in bed	TAL on H3 outlet at 175°C	TALL at 160°C risk of sulfur solidification in CR3 bed. Trip acid gas feed if TALL activates.

4.4	CR3 Temp Rise Less	Exotherm <2°C in CR3	Catalyst deactivation; very low H ₂ S/SO ₂ concentration; channeling	Near-zero conversion in CR3; 1.6 percentage points of recovery lost; tail gas H ₂ S increases	ΔT monitor across CR3	Track ΔT trend across CR3. Design exotherm is only 4.3°C any significant drop signals catalyst end-of-life.
4.5	Pressure More	Increasing ΔP across CR2/CR3	Sulfur deposition in catalyst bed; water condensation on catalyst	Bed plugging; gas bypass around catalyst; sudden ΔP spike can dislodge catalyst	ΔP indicators across CR2 and CR3	ΔP monitoring on all converter beds. High ΔP alarm. Regeneration or bed replacement on high ΔP trend.
4.6	Temp (comb) More	All three reheaters overheating	Common steam header pressure spike; simultaneous TIC failures	All converters operating above design temperature; Keq for Claus reaction decreases at all stages; significant recovery loss	Independent TICs per reheater; steam header pressure control	Independent temperature controllers per reheater do not share a single steam control valve.
4.7	Condenser C2/C3 Temp Less	C2 or C3 outlet drops below 119°C	Excess cooling; utility temperature low	Solid sulfur formation in condenser or downstream piping; blockage; pressure buildup	TIC on C2, C3 outlets	TALL at 120°C on C2, C3 outlets. Heat tracing on all sulfur drain points even if C2/C3 are near vapor-only in simulation, physical plant will have small liquid sulfur flows.
4.8	Liquid Drain No Flow	C2/C3 drains blocked	Sulfur solidification; valve failure; heat trace failure	Liquid backup into vapor stream; sulfur carryover to tail gas; OFFGAS composition change	Level indicators on C2/C3 sumps	Heat tracing critical. Even small liquid sulfur flows need reliable drainage. Do not rely on simulation vapor- only result real plant C2/C3 will drain small quantities.

Node 5: Tail Gas System (OFFGAS)

Design Intent: OFFGAS exits C3 at 130°C, 1.10 bar. Composition: 40.1 mol% N₂, 29.0 mol% H₂O, 26.9 mol% CO₂, 0.243 mol% H₂S (2.893 kmol/hr), 0.224 mol% SO₂ (2.667 kmol/hr), 2.05 mol% CO (24.42 kmol/hr), 1.19 mol% H₂ (14.15 kmol/hr), 0.298 mol% S₈. Requires downstream tail gas treatment (SCOT or equivalent) to meet >99.8% total recovery for regulatory compliance.

#	Parameter / Guide Word	Deviation	Possible Causes	Consequences	Safeguards (Industry)	Recommended Action
5.1	H ₂ S in OFFGAS More	H ₂ S above design 2.893 kmol/hr	Upstream recovery failure; catalyst deactivation; H ₂ S:SO ₂ ratio upset; reheater failure	Increased loading on tail gas treatment unit; potential breakthrough to atmosphere if TGT unit is also compromised; regulatory violation	H ₂ S analyzer on OFFGAS stream	Continuous H ₂ S analyzer on OFFGAS. High H ₂ S alarm (AAHL) triggers investigation of upstream ratio control and catalyst condition.
5.2	SO ₂ in OFFGAS More	SO ₂ above design 2.667 kmol/hr	Excess air to furnace; low H ₂ S feed; H ₂ S:SO ₂ ratio below 2:1	Regulatory SO ₂ exceedance; corrosion in tail gas treatment unit; amine absorber upset in SCOT unit	SO ₂ analyzer on OFFGAS	SO ₂ analyzer on OFFGAS with high alarm. Link to air ratio controller feedback on furnace.
5.3	CO in OFFGAS More	CO above design 24.42 kmol/hr	Furnace incomplete combustion; low air; high fuel gas rate	Increased combustible load in tail gas incinerator; CO is already significant at design any increase raises incineration duty and CO ₂ emissions	CO analyzer on OFFGAS (or tail gas incinerator inlet)	CO monitor on OFFGAS. Note CO at 24.42 kmol/hr is already substantial at design conditions furnace air ratio control is the root intervention.
5.4	S ₈ in OFFGAS More	Sulfur vapor carryover above	C3 temperature high; condenser underperformance; high sulfur vapor	Sulfur deposition in downstream piping; TGT unit fouling; tail	Temperature control on C3; online sulfur dew point	Maintain C3 at 130°C. Online sulfur dew point analyzer on OFFGAS is best

		design	pressure	gas incinerator sulfur deposition	monitoring	practice.
5.5	Temperature Less	OFFGAS below 130°C	C3 over-cooling	Sulfur condensation in OFFGAS line; potential solid sulfur plugging before TGT unit	TIC on C3 outlet	TALL at 120°C on OFFGAS line. Heat tracing on OFFGAS piping to TGT unit inlet.
5.6	H ₂ :SO ₂ Ratio Other Than	H ₂ and CO present in OFFGAS entering incinerator	Normal operation both are present at design (14.15 and 24.42 kmol/hr)	Combustible gas mixture; explosion risk if incinerator fails to light or flame-out occurs	Flame detection on tail gas incinerator; combustible gas detector	Flame detector with automatic fuel gas pilot backup on incinerator. Combustible gas detector on incinerator stack.
5.7	Total Flow No/None	No OFFGAS flow	Upstream complete blockage; process shutdown	Pressure buildup upstream; potential vessel overpressure	Pressure indicator on OFFGAS line	PAHL on OFFGAS line. Blocked outlet condition on Claus unit requires emergency shutdown of acid gas feed.
5.8	H ₂ S Leak Other Than	H ₂ S release from OFFGAS piping	Flange failure; pipe corrosion; valve leak	Toxic H ₂ S exposure (IDLH 50 ppm); fire and explosion risk with H ₂ and CO also present	Fixed H ₂ S detectors around OFFGAS piping and TGT unit; personal H ₂ S monitors for operators	Fixed H ₂ S detectors at low points and around flanges on OFFGAS system. Minimum two- detector voting for area alarm. Emergency isolation of acid gas feed on confirmed H ₂ S release.
5.9	Tail Gas Treat. No/None	TGT unit offline or bypassed	Maintenance; TGT unit upset	Untreated OFFGAS direct to incinerator or atmosphere; total sulfur recovery drops	TGT unit availability interlocked with Claus unit operation in fully	Claus unit should reduce throughput or shut down if TGT is unavailable for extended period in permitted

from >99.8% to 98.7%; regulatory exceedance	permitted plants	operation. Track H ₂ S and SO ₂ stack emissions during TGT outage.
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Summary of Critical Action Items

Priority	Action	Node	Basis
Critical	Install H ₂ S/SO ₂ ratio analyzer on process gas entering CR1 with feedback to furnace air ratio controller	Node 3	H ₂ S:SO ₂ = 2:1 is the single most important control variable. Deviation at this point propagates through all three catalytic stages.
Critical	Fixed H ₂ S detectors around furnace, OFFGAS piping, and sulfur storage	Nodes 1, 5	H ₂ S IDLH = 50 ppm. Process carries H ₂ S at all points.
Critical	TALL on all condenser outlets interlocked with heat tracing; drain line heat trace on all sulfur drainage points	Nodes 2, 4	Sulfur solidifies at 119°C. Blockage in drain lines causes sulfur backup and carryover.
High	PSV on furnace casing and condenser vessels	Nodes 1, 2	Blocked outlet scenarios create overpressure risk on all vessels.
High	ΔT monitoring across all three converter beds	Nodes 3, 4	ΔT drop from design values (47.5°C / 16.6°C / 4.3°C) is the earliest indicator of catalyst deactivation.
High	Acid gas FFLL interlock: loss of acid gas trips air supply simultaneously	Node 1	Preventing excess air in furnace without acid gas avoids explosive atmosphere formation.
Medium	Online COS analyzer on PROC-G1 if acid gas CO ₂ content is variable	Node 3	CO ₂ at 53.16 mol% is already high upstream composition changes will affect COS formation directly.
Medium	Continuous H ₂ S and SO ₂ analyzers on OFFGAS with high alarms	Node 5	Regulatory compliance monitoring. Also provides early warning of upstream upsets.

Medium	Mist eliminator on first condenser vapor outlet	Node 3	Prevents liquid sulfur entrainment into CR1, which is the most active and most valuable catalytic bed.
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Limitations and Assumptions

No formal P&ID was available. Instrument tags, line numbers, and interlock logic are generic, based on industrial Claus practice rather than a specific plant design.

The simulation uses equilibrium models throughout (RGibbs for the furnace, REquil for converters). These do not capture catalyst aging, kinetic limitations, or heat losses. Real deviations from design will be more severe than the simulation numbers suggest particularly in the later converter stages where the exotherm is already down to 4.3°C.

Condensers C2 and C3 show vapor-only outlet streams in Aspen. This is a known Flash2 convergence artifact at low sulfur concentrations, not a physical result. Real plant C2/C3 do produce small liquid sulfur flows, and this HAZOP treats them accordingly.

The waste heat boiler is merged into the RF block in the simulation. WHB-specific deviations tube rupture, steam drum level loss, cooling water fouling are assessed from standard WHB practice, not simulation data.

Safeguard recommendations are conceptually aligned with IEC 61511 SIL philosophy. Formal SIL determination requires a quantitative risk assessment, which is outside the scope of this study.

Node Boundary Definitions

Node	Description	Inlet Streams	Outlet Streams	Key Equipment
1	Reaction Furnace + Waste Heat Boiler	ACID-GAS (171.1 mol/s, 218°C, 1.77 bar); AIR (181.5 mol/s, 220°C, 1.68 bar); FUEL-GAS (3.25 mol/s, 40°C, 6.0 bar)	HOT-GAS (995.6°C, 1.5 bar)	RF (RGibbs block); WHB (implicit collapses furnace + boiler)
2	First Condenser	HOT-GAS (995.6°C, 1.5 bar)	PROC-G1 (vapor, 130°C, 1.5 bar); LS (liquid sulfur, 66.34 kmol/hr S ₂)	C (Flash2 block)
3	Reheater 1 + Catalytic Converter 1	PROC-G1 (130°C, 1.5 bar)	PROD-G1 (307.5°C, 1.34 bar); Liquid S	H1 (Heater block, 260°C outlet); CR1 (REquil

			(nil vapor only at converter)	block)
4	Reheaters 2 & 3 + Catalytic Converters 2 & 3 + Condensers C2 & C3	PROD-G1 (307.5°C, 1.34 bar)	PROC-G3 (130°C, 1.10 bar); LS1 (6.84 kmol/hr S ₈); LS2, LS3 (small vapor-only in simulation)	H2 (220°C), CR2 (236.6°C outlet); H3 (190°C), CR3 (194.3°C outlet); C2 (1.22 bar); C3 (1.10 bar)
5	Tail Gas / OFFGAS System	PROC-G3 (130°C, 1.10 bar)	OFFGAS to TGT unit or incinerator	C3 vapor outlet; OFFGAS piping; tail gas incinerator (downstream)

Guide Word Applicability by Node

Node	Parameter	Reverse	As Well As	Part Of	Notes
1 RF	Air Flow	Applied (1.4context) reverse flow into air line if furnace overpressures	Not applicable air composition is fixed atmospheric; no credible "as well as" scenario	Not applicable	Reverse flow covered under pressure deviation 1.12
1 RF	Acid Gas Comp.	Not applicable flow direction fixed by upstream pressure	Applied (1.6, 1.7) H ₂ O spike or H ₂ S drop are "other than" composition deviations	Applied (1.7) lean acid gas is partial H ₂ S content	Composition deviations are most consequential at this node
1 RF	Temperature	Not applicable	Not applicable	Applied (1.9) partial combustion at low temp is "part of" normal combustion	

2 Condenser C	Flow	Not applicable gravity-driven liquid drain; no reverse flow path without pump	Applied (2.8) water co-present in liquid sulfur drain	Not applicable	
3 CR1	Composition	Not applicable	Applied (3.6, 3.7) sulfur droplets or high COS entering CR1 alongside normal gas	Not applicable	Mist carry-over is an "as well as" deviation
4 CR2/CR3	Temperature	Not applicable	Not applicable	Applied (4.3, 4.4) partial conversion when temp is sub-optimal	
5 OFFGAS	Flow	Applied (5.7) blocked OFFGAS creates back-pressure risk	Applied (5.6) H ₂ and CO present alongside H ₂ S (combustible mixture)	Not applicable	Reverse flow not credible downstream TGT is at lower pressure

Risk Matrix

Each deviation is scored on Likelihood (1–4) and Severity (1–4). Risk = Likelihood × Severity.

Likelihood Scale:

- 1 Remote (once in plant lifetime)
- 2 Unlikely (once per decade)
- 3 Possible (once per year)
- 4 Likely (multiple times per year)

Severity Scale:

- 1 Negligible (minor process upset, no injury, no environmental impact)
- 2 Marginal (equipment damage or minor injury, recoverable)
- 3 Critical (major equipment damage, serious injury, or regulatory exceedance)
- 4 Catastrophic (fatality, major H₂S release, or plant-wide shutdown)

Risk Level:

- 1–4: Low
- 5–8: Medium
- 9–12: High
- 13–16: Critical

Dev #	Deviation	L	S	Risk Score	Risk Level
1.1	No air to furnace H ₂ S release	2	4	8	Medium
1.2	Excess air SO ₃ formation, catalyst sulfation	3	2	6	Medium
1.3	Low air H ₂ S:SO ₂ ratio upset	3	2	6	Medium
1.4	No acid gas explosive atmosphere	1	4	4	Low
1.5	Excess acid gas H ₂ S carrythrough	2	3	6	Medium
1.6	High H ₂ O in acid gas furnace quench	2	3	6	Medium
1.7	Lean acid gas flame extinction	2	4	8	Medium
1.8	Furnace overtemperature >1400°C	1	4	4	Low
1.9	Low furnace temp <900°C	2	3	6	Medium
1.10	Excess fuel gas overtemperature	2	3	6	Medium
1.11	Heavy hydrocarbon in fuel gas	2	2	4	Low
1.12	High furnace pressure reverse flow	2	3	6	Medium
1.13	Low furnace pressure H ₂ S leak	2	4	8	Medium
1.14	WHB cooling loss condenser overload	2	3	6	Medium
1.15	WHB tube rupture water ingress	1	3	3	Low
2.1	Condenser outlet <119°C sulfur solidification	3	3	9	High

2.2	Condenser outlet >130°C sulfur carryover	3	2	6	Medium
2.3	No cooling water catalyst destruction	2	4	8	Medium
2.4	High liquid sulfur level carryover	3	3	9	High
2.5	No liquid sulfur collecting	2	1	2	Low
2.6	High condenser pressure	2	2	4	Low
2.7	Drain valve stuck sulfur blockage	3	3	9	High
2.8	High H ₂ O in liquid sulfur H ₂ S at storage	2	3	6	Medium
3.1	H1 overtemperature equilibrium loss	2	2	4	Low
3.2	H1 low temperature catalyst inactive	3	3	9	High
3.3	No steam to H1 no reheating	2	3	6	Medium
3.4	CR1 outlet >320°C catalyst sintering risk	1	3	3	Low
3.5	Low ΔT across CR1 catalyst deactivation	3	3	9	High
3.6	Liquid sulfur entrainment into CR1	2	3	6	Medium
3.7	High COS loading on CR1	2	2	4	Low
3.8	H ₂ S:SO ₂ ratio off 2:1	3	3	9	High
3.9	High ΔP across CR1 bed plugging	3	2	6	Medium
4.1	CR2 inlet >220°C	2	2	4	Low
4.2	CR2 inlet <220°C low conversion	3	2	6	Medium
4.3	CR3 inlet <190°C sulfur solidification risk	2	3	6	Medium

4.4	Low ΔT across CR3 near-zero conversion	3	2	6	Medium
4.5	High ΔP in CR2/CR3 bed plugging	3	2	6	Medium
4.6	All reheaters overheat Keq loss	1	3	3	Low
4.7	C2/C3 outlet <119°C solid sulfur	3	3	9	High
4.8	C2/C3 drains blocked	3	2	6	Medium
5.1	High H ₂ S in OFFGAS	2	3	6	Medium
5.2	High SO ₂ in OFFGAS	2	3	6	Medium
5.3	High CO in OFFGAS	2	2	4	Low
5.4	High S ₈ carryover in OFFGAS	2	2	4	Low
5.5	OFFGAS below 130°C sulfur plugging	3	2	6	Medium
5.6	H ₂ /CO combustible mixture at incinerator	2	3	6	Medium
5.7	No OFFGAS flow overpressure	1	3	3	Low
5.8	H ₂ S leak from OFFGAS piping	2	4	8	Medium
5.9	TGT unit offline	2	3	6	Medium

Action Register

Action #	Description	Node	Dev Ref	Priority	Owner	Target	Status
A-01	Install air flow FFL interlock wired to simultaneous acid gas and fuel gas trip	1	1.1, 1.4	Critical	Process/Inst. Eng.	Before comm.	Open

A-02	Install O ₂ analyzer (AIC) on HOT-GAS stream with high O ₂ alarm linked to air ratio controller	1	1.2	Critical	Inst. Eng.	Before comm.	Open
A-03	Install FFIC air:acid gas ratio controller with low-flow alarm	1	1.3	Critical	Process/Inst. Eng.	Before comm.	Open
A-04	Install online feed gas analyzer (H ₂ S, CO ₂ , H ₂ O) on acid gas inlet	1	1.6, 1.7	High	Inst. Eng.	Before comm.	Open
A-05	Install TAHH on furnace at 1450°C interlocked with fuel gas and acid gas shutoff	1	1.8	High	Inst. Eng.	Before comm.	Open
A-06	Install TAL at 900°C with automatic fuel gas boost; TALL at 850°C trips acid gas feed	1	1.9	High	Process Eng.	Before comm.	Open
A-07	Install PSV on furnace casing; PAHL at 1.8 bar	1	1.12	Critical	Mech./Process Eng.	Design stage	Open
A-08	Install fixed H ₂ S detectors around furnace perimeter with PALL low pressure alarm	1	1.13	Critical	Safety/Inst. Eng.	Before comm.	Open
A-09	Install TAH on WHB outlet (>350°C) interlocked with acid gas feed trip; cooling water FFLL alarm	1	1.14	High	Process Eng.	Before comm.	Open
A-10	Install ΔP monitor across WHB with high alarm; steam drum level monitoring	1	1.15	Medium	Inst. Eng.	Before comm.	Open
A-11	Install TIC on condenser C outlet; TALL at 120°C trips cooling water supply	2	2.1	Critical	Inst. Eng.	Before comm.	Open

A-12	Install heat tracing on all sulfur drain lines from condenser C; maintain >130°C	2	2.1, 2.7	Critical	Mech. Eng.	Design stage	Open
A-13	Install LAH on liquid sulfur sump with automatic drain valve open signal	2	2.4	High	Inst. Eng.	Before comm.	Open
A-14	Install redundant cooling water pump for condenser C; FFL on cooling water	2	2.3	High	Mech. Eng.	Design stage	Open
A-15	Install PSV on condenser C vessel	2	2.6	High	Mech. Eng.	Design stage	Open
A-16	Install mist eliminator/demister pad on condenser C vapor outlet	3	3.6	Medium	Mech. Eng.	Design stage	Open
A-17	Install H ₂ S/SO ₂ ratio analyzer on PROC-G1 with feedback to furnace air ratio controller	3	3.8	Critical	Inst. Eng.	Before comm.	Open
A-18	Install ΔP indicator across CR1 with high alarm for early fouling detection	3	3.9	Medium	Inst. Eng.	Before comm.	Open
A-19	Install ΔT monitoring across all three converter beds; track trend vs design values (47.5 / 16.6 / 4.3°C)	3, 4	3.5, 4.4	High	Process Eng.	Commissioning	Open
A-20	Install FFL on steam supply to H1, H2, H3 with independent alarms	3, 4	3.3	High	Inst. Eng.	Before comm.	Open
A-21	Install TALL at 120°C on C2 and C3 condenser outlets	4	4.7	High	Inst. Eng.	Before comm.	Open
A-22	Install heat tracing on C2 and C3 sulfur drain lines;	4	4.7, 4.8	High	Mech. Eng.	Design stage	Open

LAH on C2/C3 sumps							
A-23	Install ΔP indicators on CR2 and CR3 beds with high alarm	4	4.5	Medium	Inst. Eng.	Before comm.	Open
A-24	Install independent TICs per reheater; do not share single steam control valve	4	4.6	Medium	Inst. Eng.	Design stage	Open
A-25	Install continuous H ₂ S analyzer on OFFGAS with AAHL alarm	5	5.1	High	Inst. Eng.	Before comm.	Open
A-26	Install SO ₂ analyzer on OFFGAS with high alarm linked to furnace air ratio controller	5	5.2	High	Inst. Eng.	Before comm.	Open
A-27	Install flame detector with automatic fuel gas pilot backup on tail gas incinerator	5	5.6	High	Safety/Inst. Eng.	Before comm.	Open
A-28	Install PAHL on OFFGAS line; blocked outlet trips acid gas feed	5	5.7	Medium	Inst. Eng.	Before comm.	Open
A-29	Install fixed H ₂ S detectors at low points and around flanges on OFFGAS system; minimum two-detector voting for area alarm	5	5.8	Critical	Safety Eng.	Before comm.	Open
A-30	Establish operating procedure: reduce Claus throughput or shut down if TGT unit unavailable for >4 hours in permitted operation	5	5.9	High	Process / Ops	Commissioning	Open